

Tumour Lysis Syndrome



Section I: Scenario Demographics

Scenario Title:	Tumour Lysis Syndrome (TLS)
Date of Development:	24/02/2016 (DD/MM/YYYY)
Target Learning Group:	☐ Juniors (PGY 1 - 2) ☒ Seniors (PGY ≥ 3) ☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Donika Orlich (modified case Dysrhythmia secondary to hyperkalemia by Kyla Caners)
Affiliations/Institution(s):	McMaster University
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To expose learners to a case of Tumour Lysis Syndrome with severe hyperkalemia, hyperuricemia and hypocalcemia as its metabolic complications.
CRM Objectives:	1) Demonstrate leadership by maintaining a global perspective and organizing team to deliver multiple drug therapies 2) Utilize resources appropriately by calling for help when indicated 3) Demonstrate problem solving ability by implementing concurrent management approach and avoiding fixation error 4) Employ communication skills to obtain an immediate code status from a family member
Medical Objectives:	1) Recognize hyperkalemia on ECG and initiate ED therapy 2) Investigate for the possibility of tumour lysis syndrome in the setting of recent chemotherapy for hematologic cancer by highlighting the importance of serum electrolyte levels and their disturbances, including hyperuricemia. 3) Considers indications for treatment of hypocalcemia in the ED setting 4) Consults nephrology for dialysis

Case Summary: Brief Summary of Case Progression and Major Events

A 72-year-old male is brought in as a "code STEMI" to the resuscitation bay. He was recently diagnosed with ALL and had chemotherapy 3 days ago for the first time. The patient is severely hyperkalemic, which must be initially recognized and treated, hypocalcemic and hyperuricemic as a result of Tumour Lysis Syndrome and the metabolic derangements must be stabilized until emergent hemodialysis is arranged.

References

- Pfennig CL, Slovis CM. (2013). Electrolyte disorders. In J. Marx, R. Hockberger & R. Walls (Eds.), Rosen's emergency medicine - concepts and clinical practice. pp. (1636-51). Philadelphia, PA.:Saunders.
- Website: accessed on April 14, 2015: <http://lifeinthefastlane.com/ecg-library/basics/hyperkalaemia/>
- Website: accessed on Feb 24, 2016: <http://lifeinthefastlane.com/cc/tumour-lysis-syndrome/>



Tumour Lysis Syndrome



Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☒ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
Confederates		Brief Description of Role	
Wife:	The wife arrives with the patient. The wife will stress the fact that he would not want CPR or intubation unless the acute condition is reversible. She will ask the team leader directly "is this a reversible problem?". The wife will then ask general questions ("what's happening now?") and will follow directions if asked to leave the bedside.		
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☒ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
None required.			
E. Approximate Timing			
Set-Up:	3 min	Scenario:	15 min
		Debriefing:	30 min

Tumour Lysis Syndrome



Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case					
72M complaining of general weakness for 2 days. His wife called EMS and he was a STEMI patch to your hospital. He has been placed in the resuscitation bay.					
B. Patient Profile and History					
Patient Name: Geoff Brady		Age: 72		Weight: 80kg	
Gender: ☒ M ☐ F		Code Status: DNR			
Chief Complaint: Weak and dizzy					
History of Presenting Illness: Felt unwell since waking up this morning. Feels light headed with standing and like he "has no energy." Nauseous. Has newly diagnosed ALL and had his first round of chemotherapy 3 days ago.					
Past Medical History:	ALL		Medications:	Ramipril 5mg OD	
	HTN			Crestor 20mg OD	
	Dyslipidemia			Cisplatin	
Allergies: Penicillin					
Social History: Retired. Lives with his wife. No EtOH/smoking/illicit drug use.					
Family History: No hx CAD. All have diabetes.					
Review of Systems:	CNS:	Generally weak.			
	HEENT:	Nil.			
	CVS:	No CP. No palps.			
	RESP:	No SOB.			
	GI:	Nausea.			
	GU:	Nil			
	MSK:	Muscle cramps and paresthesia's	INT:	Nil.	
C. Baseline Simulator State and Physical Exam					
☐ No Monitor Display		☒ Monitor On, no data displayed		☐ Monitor on Standard Display	
HR: 50/min	BP: 92/65	RR: 16/min	O ₂ SAT: 97% RA		
Rhythm: Idioventricular	T: 36.4°C	Glucose: 12.7 mmol/L	GCS: 13 (E3 V4 M6)		
General Status: Slightly diaphoretic. Looks unwell.					
CNS:	A+Ox3. No focal neurologic deficits. No asterixis.				
HEENT:	PERLA. 3mm.				
CVS:	Normal S1/S2. No murmur.				
RESP:	GAEB. No adventitious sounds.				
ABDO:	Soft, NT.				
GU:	Nil.				
MSK:	Nil.	SKIN:	No rashes.		

Tumour Lysis Syndrome



Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: idioventricular HR: 50/min BP: 92/65 RR: 20/min O ₂ SAT: 97% RA T: 36.4°C	Diaphoretic, awake, answering questions. Feels dizzy.	<u>Learner Actions</u> ☐ IV, O ₂ , monitors ☐ ECG ☐ Blood work including trop, extended lytes, uric acid, lactate ☐ Fluid bolus ☐ Check cap sugar (12.7) ☐ Administer calcium gluconate 1g (10-30ml) iv ☐ ± ASA 160mg PO chew	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - Calcium given → HR 70, QRS narrows (change to sinus) - Fluid bolus → BP 102/67 <u>Triggers</u> <i>For progression to next state</i> - Calcium not given by 3 min → 2. Arrest - Initial work-up complete and calcium given → 4. VBG Back
2. Arrest Rhythm → Asystole BP → -/- RR → 0 O ₂ SAT → ?	Unresponsive with no vitals.	<u>Learner Actions</u> ☐ Quality CPR ☐ Calcium chloride 1-2 amps ☐ NaHCO ₃ amps ☐ Epinephrine amps ☐ 1 amp D50 then insulin R 10 units iv	<u>Modifiers</u> - 7 minutes into case → critical VBG back showing hyperkalemia <u>Triggers</u> - Calcium given → 3. NSR
3. NSR Rhythm → sinus HR → 70 BP → 105/70 RR → 16 O ₂ SAT → 97% RA	Awake and responding.	<u>Learner Actions</u> ☐ Ventolin 20mg nebulized, 24 puffs, or 0.5mg IV ☐ 1amp D50 then insulin R 10 units iv (if not yet done) ☐ Repeat ECG ☐ ± HCO ₃ amp ☐ Call nephro re: dialysis	<u>Modifiers</u> - Ventolin given → HR 90 <u>Triggers</u> - Two treatments for hyperK given and Nephro called → 4. Blood Work Back - 12 minutes → 4. Blood Work Back
4. VBG Back Rhythm → sinus HR → 70 BP → 105/70	Awake, alert, no longer feels dizzy.	<u>Learner Actions</u> ☐ Ventolin 20mg nebulized, 24 puffs, or 0.5mg IV ☐ 1amp D50 then insulin R 10 units iv ☐ Repeat ECG ☐ ± HCO ₃ amp ☐ ± Lasix 40mg iv (if still makes urine) ☐ Call nephro re: dialysis	<u>Modifiers</u> - Given critical VBG result showing hyperkalemia as state starts - Ventolin given → HR 90 <u>Triggers</u> - Two treatments for hyperK given and Nephro called → 4. Blood Work Back - 12 minutes → 4. Blood Work Back

Tumour Lysis Syndrome



5. Blood Work Back	Patient still complaining of muscle cramps, otherwise better.	<u>Learner Actions</u> - ☐ Recognize likely Tumour Lysis syndrome - ☐ IV NV bolus (goal u/o 2cc/h) - ☐ Allopurinol (600-800mg/d) - ☐ Rasburicase - ☐ ± Lasix 20mg IV (hyperK, volume overload, poor u/o) - ☐ IV Calcium gluconate (1-2g over 10 minutes) - ☐ Call MedOnc re admission - ☐ Call Nephro for dialysis consideration	<u>Modifiers</u> - Given lab results showing hyperuricemia, hyperphosphatemia and hypocalcemia as state starts - If no treatment for hyperuricemia initiated, phone call from the "Lab" informing that the "<u>Urate and calcium at a level that must be treated</u>", <u>Triggers</u> - Considers or initiates a treatment for hyperuricemia and hypocalcemia given AND Nephro called for urgent dialysis-> END CASE *Consideration of treatment includes voicing that hem/onc will be consulted for guidance in treatment.
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Tumour Lysis Syndrome



Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results	
Critical VBG: pH 7.27 PCO2 35 PO2 45 HCO3 20 Lactate 2.5 Na 142 K 8.6 Cl 105	Labs: Troponin: 5 INR: 1.0 WBC: 8 Neuts: 1.0 Hb: 140 Plt: 300 Urate 400 umol/L (high) Cr: 650 umol/L (high) PO4: 3.0 mmol/L (high) Ca: 1.98 mmol/L (low)
Images (ECGs, CXRs, etc.)	
ECG 1: Hyperkalemia STEMI mimic (Source: http://lifeinthefastlane.com/ecg-library/basics/hyperkalaemia/)	
ECG 2: NSR with no ST changes (Source: http://cdn.lifeinthefastlane.com/wp-content/uploads/2011/12/normal-sinus-rhythm.jpg)	

Tumour Lysis Syndrome



Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) What are the ECG changes associated with hyperkalemia? 2) What are some mimics of ST elevation on an ECG? 3) What lab abnormalities can be seen with tumour lysis syndrome? What causes each? 4) What are symptoms of hypocalcemia? What is the risk in treating it?	
Key Moments	
Identification of hyperkalemia on ECG.	
Alteration of ACLS management to include copious calcium and HCO ₃ in context of hyperkalemia	
Recognition of Tumor Lysis Syndrome as the cause of hyperkalemia and initiation of therapy aimed at hyperuricemia and symptomatic hypocalcemia.	